

STAT 5309 – SPRING 2019

LAB 6

****Contents:** 2^K (Two-level factors) and 3^K (Three -level factors) Factorial design

DUE: Sun, Mar 31

A. PRACTICE

##-----2-level factors-----

-----Example of 2^3 Full Factorial Design: 3 factors. Each with 2 levels.

The 2 levels are denoted by (-) [low level] and (+) [high]. The two levels need to be spread out in order to obtain a larger range in the response.

```
install.packages("FrF2")
library(FrF2)
```

```
install.packages("daewr")
library(daewr)
```

volt # from daewr

	A	B	C	y
1	22	0.5	0.5	705
2	32	0.5	0.5	620
3	22	5	0.5	700
4	32	5	0.5	629
5	22	0.5	5	672
6	32	0.5	5	668
7	22	5	5	715
8	32	5	5	647
9	22	0.5	0.5	680
10	32	0.5	0.5	651
11	22	5	0.5	685
12	32	5	0.5	635
13	22	0.5	5	654
14	32	0.5	5	691
15	22	5	5	672
16	32	5	5	673

```
volt.lm<- lm(y ~ A*B*C, data=volt, contrast=list(A=contr.FrF2, B=contr.FrF2, C=contr.FrF2) )
```

```
summary(mod.1)
```

Call:

```
lm.default(formula = y ~ A * B * C, data = volt, contrasts = list
(A = contr.FrF2,
  B = contr.FrF2, C = contr.FrF2))
```

Residuals:

Min	1Q	Median	3Q	Max
-21.50	-11.75	0.00	11.75	21.50

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
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```

(Intercept) 668.5625    4.5178 147.985 4.86e-15 ***
A1          -16.8125    4.5178  -3.721 0.00586 **
B1           0.9375    4.5178   0.208 0.84079
C1           5.4375    4.5178   1.204 0.26315
A1:B1       -6.6875    4.5178  -1.480 0.17707
A1:C1       12.5625    4.5178   2.781 0.02390 *
B1:C1        1.8125    4.5178   0.401 0.69878
A1:B1:C1     -5.8125    4.5178  -1.287 0.23422
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

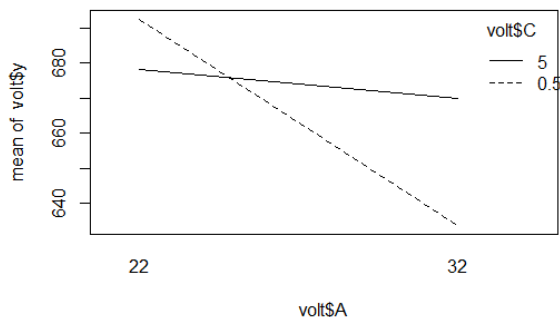
```

Residual standard error: 18.07 on 8 degrees of freedom
 Multiple R-squared: 0.772, Adjusted R-squared: 0.5724
 F-statistic: 3.869 on 7 and 8 DF, p-value: 0.0385

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##-----PLOTS: interactions

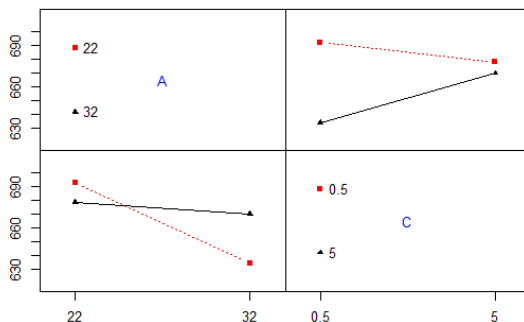
interaction.plot(volt\$A, volt\$C, volt\$y)



IAPlot(mod_1, select=c(1,3))

#package FrF2

Interaction plot matrix for y



Note: One of the interaction plot at the lower left indicates INTERACTION

##-----2-level Full Factorial design-----

##-----Example of 2^4 Full Factorial Design: 4 factors; Each has 2 levels.

data(chem) # from daewr

	A	B	C	D	y
1	-1	-1	-1	-1	45
2	1	-1	-1	-1	41
3	-1	1	-1	-1	90
4	1	1	-1	-1	67
5	-1	-1	1	-1	50
6	1	-1	1	-1	39
7	-1	1	1	-1	95
8	1	1	1	-1	66
9	-1	-1	-1	1	47
10	1	-1	-1	1	43
11	-1	1	-1	1	95
12	1	1	-1	1	69
13	-1	-1	1	1	40
14	1	-1	1	1	51
15	-1	1	1	1	87
16	1	1	1	1	72

```
chem.lm<- lm(y ~ A*B*C*D, data=chem)
summary(chem.lm)
```

Call:

```
lm.default(formula = y ~ A * B * C * D, data = chem)
```

Residuals:

ALL 16 residuals are 0: no residual degrees of freedom!

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	62.3125	NA	NA	NA
A	-6.3125	NA	NA	NA
B	17.8125	NA	NA	NA
C	0.1875	NA	NA	NA
D	0.6875	NA	NA	NA
A:B	-5.3125	NA	NA	NA
A:C	0.8125	NA	NA	NA
B:C	-0.3125	NA	NA	NA
A:D	2.0625	NA	NA	NA
B:D	-0.0625	NA	NA	NA
C:D	-0.6875	NA	NA	NA
A:B:C	-0.1875	NA	NA	NA
A:B:D	-0.6875	NA	NA	NA
A:C:D	2.4375	NA	NA	NA
B:C:D	-0.4375	NA	NA	NA
A:B:C:D	-0.3125	NA	NA	NA

Residual standard error: NaN on 0 degrees of freedom
Multiple R-squared: 1, Adjusted R-squared: NaN
F-statistic: NaN on 15 and 0 DF, p-value: NA

Notes:

P-values can't be calculated using single replicate.

Can use Tukey1() for single replicate

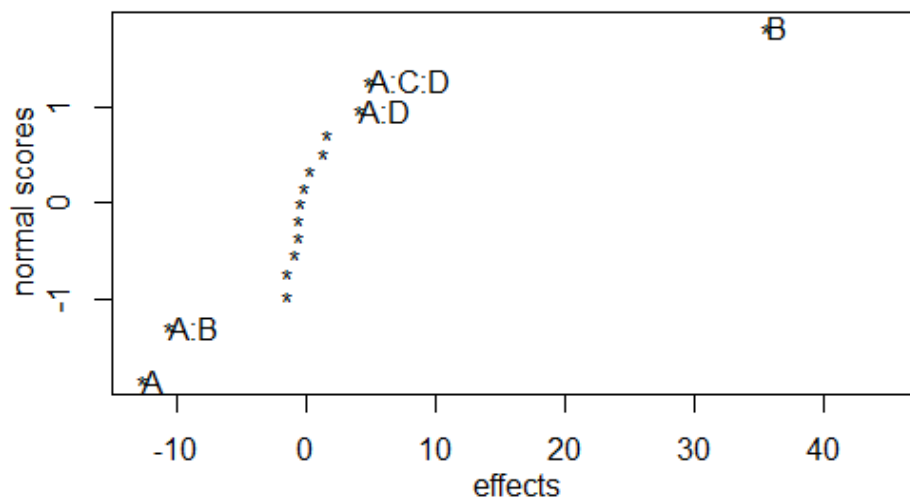
#####-----PLOTS -----

- (1) Normal Probability/Half Normal Plots of Effects [Daniel Plot, Lenth Plot]: DanielPlot(), LenthPlot(), fullnormal()
- (2) Main Effects Plot: MEPlot()
- (3) Interaction Plot: IAPlot()

DanielPlot(mod.2)

#package FrF2

Normal Plot for y, alpha=0.05



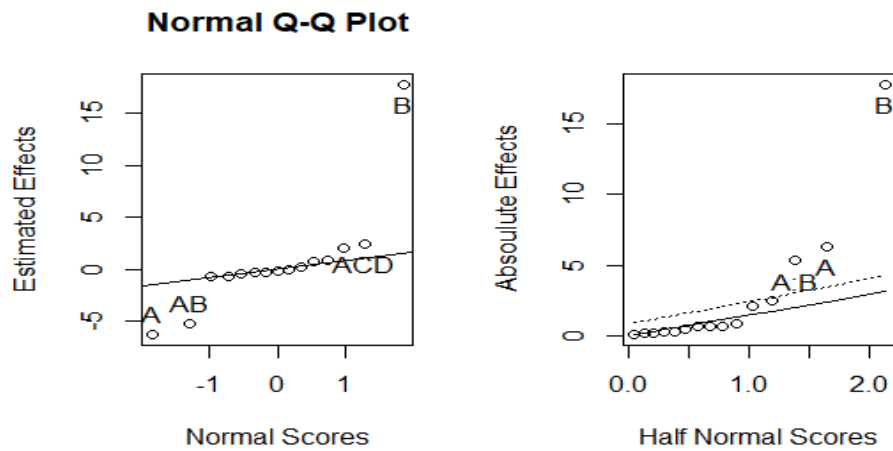
Note: If effects are off the straight line , then they are SIGNIFICANT

par(mfrow=c(1,2))

fullnormal(coef(mod.1)[-1])

LGB(coef(mod.1)[-1])

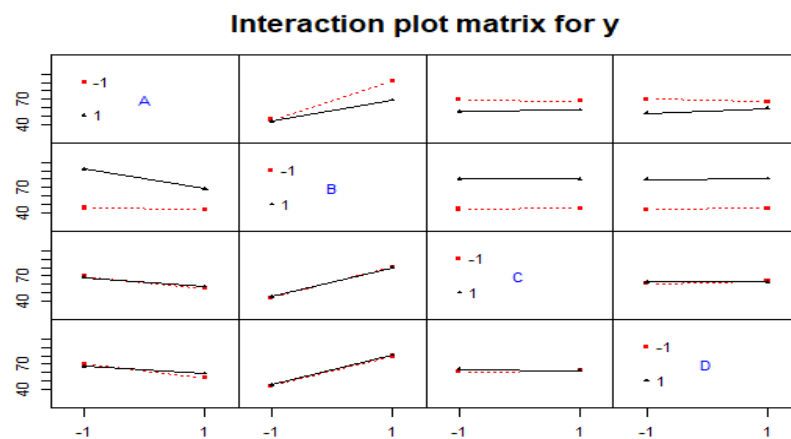
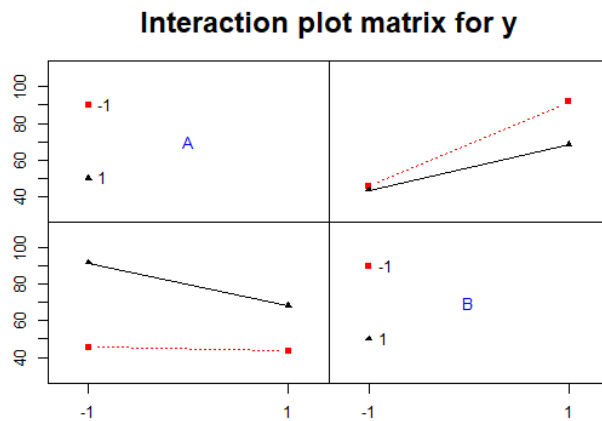
#package daewr



#Note: significant interaction AB, ACD

```
IAPlot(mod.1, select=c(1,2))
IAPlot(mod.1, select=c(1,3,4))
```

package FrF2



2^K or 3^K – Designs

```
## expand.grid();
```

```
D <- expand.grid(BW=c(3.25, 3.75, 4.25), WL=c(4,5,6))  
# generate a  $3^2$  (2 – factors, 3 – levels); total is 9 runs ( $3^2$ ). Response not included
```

	BW	WL
1	3.25	4
2	3.75	4
3	4.25	4
4	3.25	5
5	3.75	5
6	4.25	5
7	3.25	6
8	3.75	6
9	4.25	6

```
##gen.factorial(); Package “AlgDesign” : Generate a full factoriral design
```

```
library(AlgDesign)
```

```
D.1 <- gen.factorial(levels=c(2,2,2),nVars=3, center=T, varNames=c("A", "B", "C"))
```

```
D.1 #generate 3 factors, each with 2 levels, with names of factors;  
# total is 8 runs ( $2^3$ )
```

```
y <- c(1.5, 2.0, 2.1, 3.0, 1.2, 2.1, 2.5, 2.8)
```

```
data <- cbind(D.1,y)
```

```
data
```

	A	B	C	y
1	-1	-1	-1	1.5
2	1	-1	-1	2.0
3	-1	1	-1	2.1
4	1	1	-1	3.0
5	-1	-1	1	1.2
6	1	-1	1	2.1
7	-1	1	1	2.5
8	1	1	1	2.8

```
mod.3 <- lm(y ~ A*B*C, data=data)
```

```
summary(mod.3)
```

```
Call:
```

```
lm.default(formula = y ~ A * B * C, data = data)
```

```
Residuals:
```

ALL 8 residuals are 0: **no residual degrees of freedom!**

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.15e+00	NA	NA	NA
A	3.25e-01	NA	NA	NA
B	4.50e-01	NA	NA	NA
C	1.57e-16	NA	NA	NA
A:B	-2.50e-02	NA	NA	NA
A:C	-2.50e-02	NA	NA	NA
B:C	5.00e-02	NA	NA	NA
A:B:C	-1.25e-01	NA	NA	NA

Residual standard error: NaN on 0 degrees of freedom

Multiple R-squared: 1, Adjusted R-squared: NaN

F-statistic: NaN on 7 and 0 DF, p-value: NA

> `anova(mod.3)`

Analysis of Variance Table

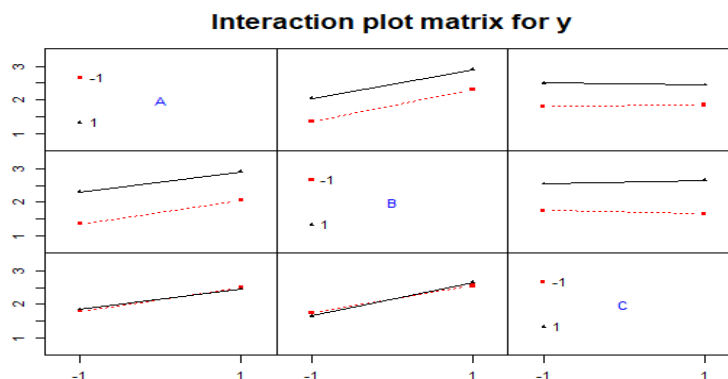
Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
A	1	0.845	0.845		
B	1	1.620	1.620		
C	1	0.000	0.000		
A:B	1	0.005	0.005		
A:C	1	0.005	0.005		
B:C	1	0.020	0.020		
A:B:C	1	0.125	0.125		
Residuals	0	0.000			

➔ **Note: Single replicate causes P-values not calculated.**

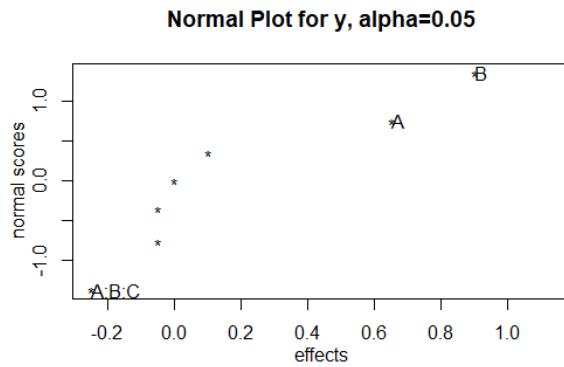
#-----Interaction-----

IAPlot(mod_3, select=c(1,2, 3))



#----- Half Normal Plot of Effects-----

DanielPlot(mod.3)



B. EXERCISE

1.

A router is used to cut locating notches on a printed circuit board. The vibration level at the surface of the board as it is cut is considered to be a major source of dimensional variation in the notches. Two factors are thought to influence vibration: bit size (A) and cutting speed (B). Two bit sizes ($\frac{1}{16}$ and $\frac{1}{8}$ inch) and two speeds (40 and 90 rpm) are selected, and four boards are cut at each set of conditions shown below. The response variable is vibration measured as the resultant vector of three accelerometers (x , y , and z) on each test circuit board.

A	B	Treatment Combination	Replicate			
			I	II	III	IV
—	—	(1)	18.2	18.9	12.9	14.4
+	—	<i>a</i>	27.2	24.0	22.4	22.5
—	+	<i>b</i>	15.9	14.5	15.1	14.2
+	+	<i>ab</i>	41.0	43.9	36.3	39.9

- Analyze the data from this experiment.
- Construct a normal probability plot of the residuals, and plot the residuals versus the predicted vibration level. Interpret these plots.
- Draw the AB interaction plot. Interpret this plot. What levels of bit size and speed would you recommend for routine operation?

2.

An engineer is interested in the effects of cutting speed (A), tool geometry (B), and cutting angle (C) on the life (in hours) of a machine tool. Two levels of each factor are chosen, and three replicates of a 2^3 factorial design are run. The results follow:

A	B	C	Treatment Combination	Replicate		
				I	II	III
–	–	–	(1)	22	31	25
+	–	–	a	32	43	29
–	+	–	b	35	34	50
+	+	–	ab	55	47	46
–	–	+	c	44	45	38
+	–	+	ac	40	37	36
–	+	+	bc	60	50	54
+	+	+	abc	39	41	47

- Estimate the factor effects. Which effects appear to be large?
- Use the analysis of variance to confirm your conclusions for part (a).
- Write down a regression model for predicting tool life (in hours) based on the results of this experiment.
- Analyze the residuals. Are there any obvious problems?
- Based on an analysis of main effect and interaction plots, what levels of A , B , and C would you recommend using?

3.

An experiment was performed to improve the yield of a chemical process. Four factors were selected, and two replicates of a completely randomized experiment were run. The results are shown in the following table:

Treatment Combination	Replicate		Treatment Combination	Replicate	
	I	II		I	II
(1)	90	93	d	98	95
a	74	78	ad	72	76
b	81	85	bd	87	83
ab	83	80	abd	85	86
c	77	78	cd	99	90
ac	81	80	acd	79	75
bc	88	82	bcd	87	84
abc	73	70	$abcd$	80	80

- (a) Estimate the factor effects.
- (b) Prepare an analysis of variance table, and determine which factors are important in explaining yield.
- (c) Write down a regression model for predicting yield, assuming that all four factors were varied over the range from -1 to $+1$ (in coded units).
- (d) Plot the residuals versus the predicted yield and on a normal probability scale. Does the residual analysis appear satisfactory?
- (e) Two three-factor interactions, ABC and ABD , apparently have large effects. Draw a cube plot in the factors A , B , and C with the average yields shown at each corner. Repeat using the factors A , B , and D . Do these two plots aid in data interpretation? Where would you recommend that the process be run with respect to the four variables?